



Basilisk

PRODUCT REQUIREMENTS DOCUMENT

A Cardano On-Chain Analytics, Portfolio & Trading Platform

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1. Executive Summary

Basilisk is an all-in-one analytics, portfolio-tracking, and trading platform for the Cardano blockchain. It targets the same job-to-be-done as **TapTools** — today's market leader — while differentiating on speed, price transparency, real-time data, trader-grade tooling, and an open developer API.

The name nods to the **basilisk** — the serpent-king of myth — keeping the Cardano "snek" lineage while evoking **Roko's Basilisk**, the AI thought experiment: a fitting emblem for an agent-native platform. The product spans three surfaces:

- **Web app** — charts, portfolio, wallet profiling, market screeners, DeFi/NFT analytics, and in-app swaps.
- **Public Data API** — a paid REST + WebSocket API for developers, bots, and integrators.
- **Mobile app + alerts** — portfolio tracking, price/whale alerts, and push notifications.

Crucially, the API is **agent-native**: pay-per-call via the **x402** micropayment protocol (no keys, no subscription) and queryable by LLM agents through an **MCP server** — turning Basilisk into the data and execution backbone for **agent-to-agent (A2A) trading** on Cardano (see §8).

THE OPPORTUNITY

TapTools is dominant but carries known friction: an **NFT-gated subscription model** that confuses non-crypto-native users, occasional data latency, and an API priced for enterprises. Basilisk can win the **prosumer trader + builder** segment with cleaner UX, real-time streaming data, transparent fiat pricing, and a generous free developer tier.

3

Product surfaces: web, API, mobile

<5s

On-chain data freshness target

5+

DEX adapters at MVP (Minswap, Sundae, WingRiders, Muesli, VyFi)

12mo

To full-scale platform across 4 phases

2. Background & Market Research

2.1 What TapTools Is

TapTools is the #1 token, NFT, and wallet tracking platform on Cardano. Its core surfaces define the parity bar Basilisk must meet or beat:

Area	Capabilities
Portfolio tracker	Connect/watch multiple wallets; net worth in ADA/USD; value-over-time chart; holdings (tokens, NFTs, LP positions, debt/loans); trade history; P&L.
Token charts	Price charts, last ~500 trades, token info, market-sentiment indicators, OHLCV.
Market screeners	Top 50 tokens, Top Gainers/Losers, volume leaders, trending.
Wallet Profiler PRO	Deep-dive on any wallet: total & liquid value, holdings breakdown, volume, activity patterns.
Hot Wallets / smart money	Track high-activity & profitable wallets ("whale" watching).
Token Distribution	Holder concentration, top holders, supply distribution — for due diligence.
DEX analytics	TVL, APYs, liquidity across Minswap, SundaeSwap, WingRiders, MuesliSwap, VyFinance, etc.
NFT analytics	Collection stats, floor, sales history, listings, distribution.
In-app swaps	Swap aggregation so users don't leave the app.
Alerts	Wallet / balance / profit-change alerts.
Public API	Tiered REST API for tokens, NFTs, market, wallet, on-chain data.

2.2 TapTools Business Model

- **PRO is NFT-gated.** "Tappy" NFTs grant access — e.g. a 4,000-supply sale at **420 ADA each**; lifetime via 5 original Tappy NFTs; annual via a single Tappy02 NFT. Secondary-market pricing floats after the

primary sale.

- **Day passes:** ~10 ADA for a 1-day PRO pass.
- **API:** separate developer subscription tiers; positioned as one of the most in-depth API services in the ecosystem.
- **Quality thresholds:** Top 50 requires ≥1M ADA lifetime volume; Top Gainers requires ≥5,000 ADA LP; NFT portfolio requires ≥5,000 ADA lifetime volume.

2.3 API Endpoint Surface (Parity Reference)

The TapTools API is organized into the categories below; Basilisk should match or exceed each:

Category	Endpoints / Data
Token	price/quote, top tokens, market cap, volume stats, holder count, top holders, OHLCV, trades, price-change, links/metadata.
NFT	collection stats, floor, trades, listings, distribution, asset sales history, traits/rarity.
Market	aggregated stats: total volume, active addresses, holders, market-wide metrics.
Wallet	portfolio positions, token/NFT holdings, LP positions, trade history, value-over-time.
On-chain / Integrations	asset supply, block info, events, DEX pairs, protocol data.

2.4 Data Sources & How It Works

TapTools aggregates from Cardano DEXs (Minswap, SundaeSwap, WingRiders, MuesliSwap, VyFinance, Spectrum) and on-chain data — commonly paired with [Blockfrost](#) for chain access. Prices derive from DEX liquidity-pool reserves and swap transactions; portfolio valuation marks holdings to current pool prices.

2.5 Competitive Landscape

Product	Focus	Notes
TapTools	Analytics + portfolio + API	Market leader; NFT-gated PRO.
DexHunter	DEX aggregator	Routes the majority of Cardano on-chain swaps; strong execution + analytics.
Cardanoscan	Block explorer	Transactions, wallets, stake pools, assets.
Cexplorer	Explorer / analytics	By Vellum Labs; epochs, pools, assets.
ADASat / pool.pm	Explorer / asset viz	Independent explorers; pool.pm for visual asset/wallet browsing.

THE GAP BASILISK EXPLOITS

Real-time streaming data, transparent fiat + crypto pricing (not NFT-gated), TradingView-class charting, a generous developer API, best-in-class smart-money / whale tooling, and deeper DeFi coverage (lending, liquid staking, yield).

3. Goals & Non-Goals

3.1 Goals

1. Achieve **feature parity** with TapTools' core analytics + portfolio surfaces by MVP.
2. Differentiate on **real-time WebSocket data**, **fiat-transparent pricing**, and **trader UX**.
3. Ship a **developer API with a free tier** to seed an ecosystem of bots and integrations.
4. Deliver sub-second chart loads and **<5s on-chain data freshness**.
5. Build a sustainable revenue model: subscriptions + API + optional swap fees.

3.2 Non-Goals (v1)

- ✗ Building our own DEX or AMM — we **aggregate**, not operate liquidity.
- ✗ Cross-chain support beyond Cardano (future consideration).
- ✗ Custodial wallet / fund custody — Basilisk is non-custodial, read-only + sign-via-wallet.
- ✗ A full tax-reporting engine (post-MVP integration / partner).

4. Target Users & Personas

Persona	Needs	Key Features
Active Trader "Degen Dan"	Fast charts, real-time prices, new-token discovery, swap execution, whale alerts.	Charts, screeners, in-app swap, alerts, smart-money.
Long-Term Holder "HODL Hana"	Portfolio net worth, P&L, staking/yield tracking, simple alerts.	Portfolio, DeFi positions, alerts.
Project / SPO "Builder Bo"	Token distribution, holder analytics, liquidity health, embeddable widgets.	Token distribution, holder analytics, API.
Developer / Bot "Dev Dee"	Reliable price/market/wallet API, WebSocket streams, great docs.	Public API, SDKs, webhooks.
NFT Collector "Collector Cleo"	Floor tracking, collection analytics, rarity, NFT portfolio value.	NFT analytics, portfolio.

5. Product Scope & Features

5.1 MVP Phase 1

Feature	Description
A. Portfolio Tracker	CIP-30 wallet connect (Eternl, Lace, Nami, Vespr, Typhon) read-only + optional sign-to-swap; multi-wallet & watch-only; net worth in ADA + USD + user fiat; value-over-time; holdings across tokens, NFTs, LP, liquid-staking, lending; realized/unrealized P&L; trade history.
B. Token Charts	TradingView-class candlesticks; OHLCV across timeframes; live DEX-aggregated (VWAP) price; streaming trades feed; token info (supply, mcap, FDV, liquidity, holders, links, policy ID); price-change, ATH/ATL.
C. Market Screeners	Top tokens by mcap/volume; Gainers/Losers; Trending; New listings; filters (liquidity, volume, age) with quality thresholds to suppress spam.
D. Wallet Profiler / Smart Money	Look up any address (total/liquid value, holdings, volume, activity, P&L); curated smart-money lists; follow wallets; whale-trade feed.
E. Token Distribution	Top holders, concentration (Gini / top-10 %), supply distribution, LP-vs-wallet split.
F. Alerts	Price thresholds, % moves, whale buys/sells, balance changes, new LP/listing — via in-app, push, email, Telegram/Discord webhook.
G. Public Data API (Beta)	REST endpoints mirroring §2.3; API keys; rate-limited free tier.

5.2 Phase 2 Trader + API

- **In-app swaps** via DEX-aggregator routing (best price across Minswap/SundaeSwap/WingRiders/...), non-custodial signing.
- **WebSocket streaming API** (live prices, trades, blocks).
- **DeFi dashboard**: TVL, APYs, yield/lending/liquid-staking positions across protocols.
- **NFT analytics suite**: floors, sales, listings, rarity, collection screeners.
- **Mobile apps** (iOS/Android) on React Native / Expo.
- **Embeddable widgets** (price cards, charts) for projects.

5.3 Phase 3+ Scale & Moat

- Copy-trading / strategy automation; bot marketplace.
- Tax/accounting export & partner integrations.
- Advanced on-chain forensics (fund-flow graphs, cluster labeling).
- AI assistant — natural-language queries over on-chain data via an MCP server.
- Governance / Catalyst & staking analytics.

6. Differentiation Strategy

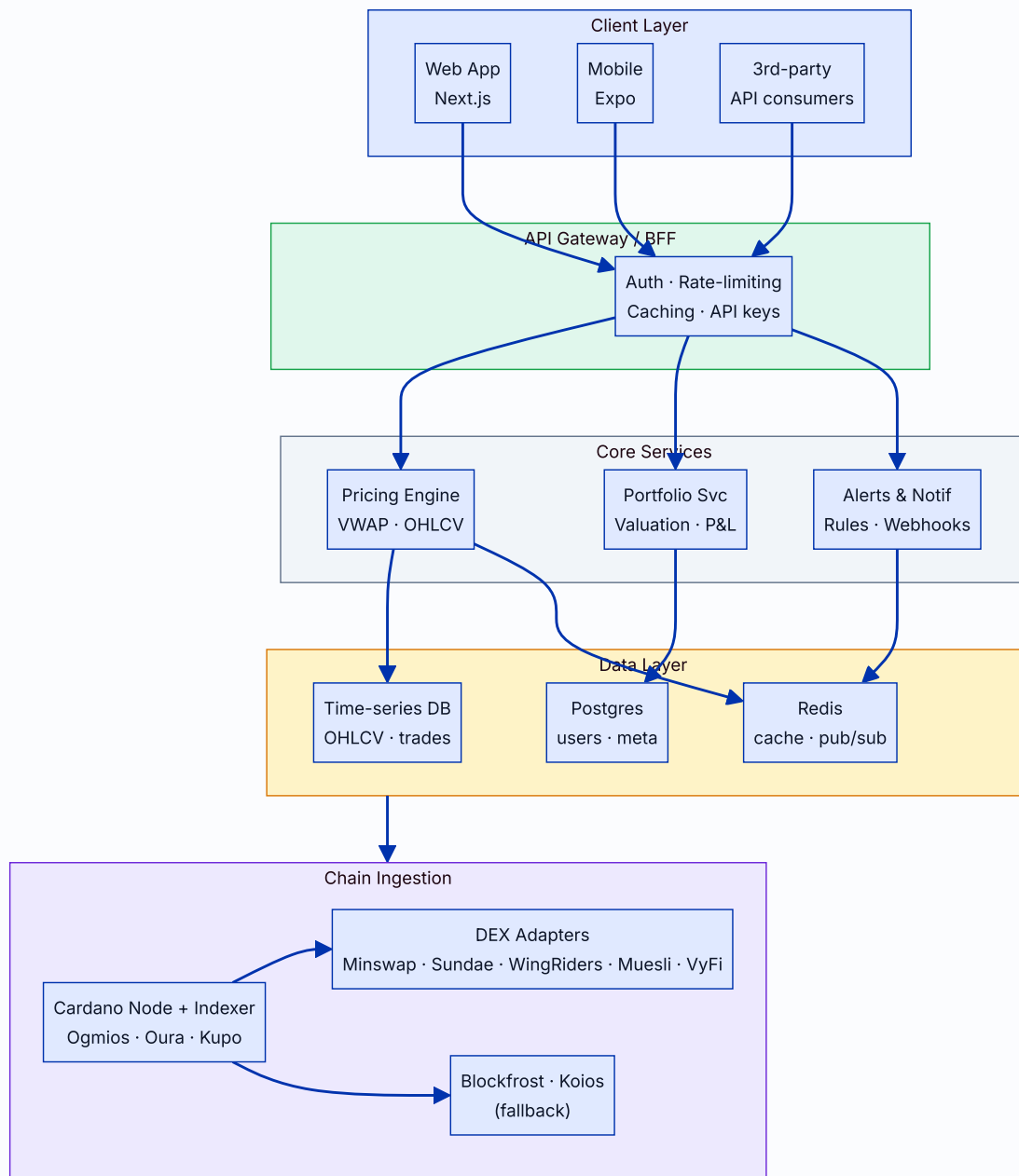
Lever	TapTools Today	Basilisk
Pricing	NFT-gated PRO (confusing for new users)	Transparent fiat + crypto subscriptions; NFT/token = perk, not requirement.
Data freshness	Periodic refresh; some latency	Real-time WebSocket streaming; <5s freshness target.
Developer API	Premium, enterprise-leaning	Generous free tier + clean docs + SDKs to seed the ecosystem.
Charting	Good	TradingView-grade with drawing tools & indicators.
Smart money	Hot wallets	Labeled clusters, follow + auto-alerts, PnL leaderboards.
DeFi depth	DEX / TVL focus	Lending, liquid staking, yield positions unified in portfolio.
Onboarding	Crypto-native	Email/social login option + watch-only first; wallet connect optional.
Agent access	Subscription + API-key gated	x402 pay-per-call (no keys/signup) + MCP server — built for autonomous agents.

7. Technical Architecture

The hard part isn't the UI — it's the data pipeline. Three layers do the heavy lifting: **chain ingestion**, the **pricing engine**, and the **real-time API**.

7.1 High-Level Architecture

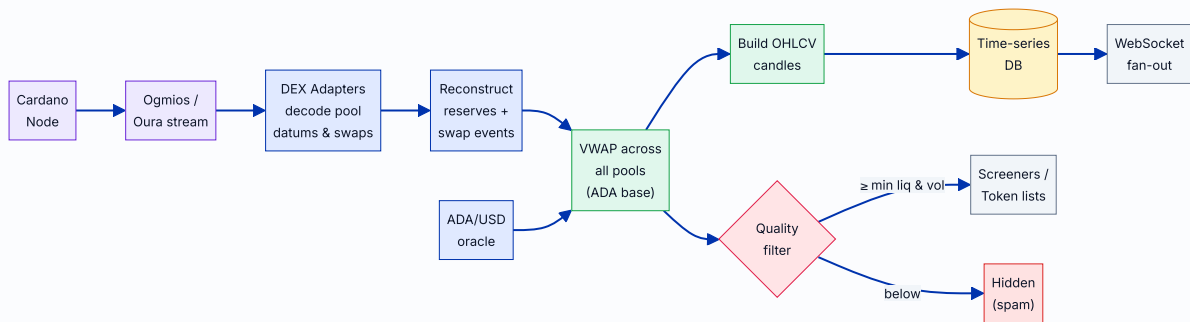
FIGURE 7.1 — SYSTEM OVERVIEW



Clients reach core services through a single gateway; services read from a shared data layer fed by the chain-ingestion tier.

7.2 Chain Ingestion & Pricing Pipeline

FIGURE 7.2 — FROM BLOCK TO PRICE



DEX adapters are the highest-value, highest-maintenance work — each protocol represents pools differently on-chain.

7.3 Component Responsibilities

Layer	Responsibility & Candidate Stack
Chain ingestion	Cardano node + chain-follower: Ogmios (node WS bridge) + Oura (event stream) for real-time; Kupo / Carp for queryable history; Blockfrost + Koios as managed fallback / cross-check.
DEX adapters	Decode pool scripts/datums per protocol to reconstruct reserves, swaps, LP mint/burn → derive price & liquidity. The core moat.
Pricing engine	Volume-weighted price across pools (ADA base + ADA/USD oracle); OHLCV candles persisted in TimescaleDB / ClickHouse; spam/quality filters.
Portfolio service	Resolve UTxOs → native-asset balances, NFTs, LP tokens (decoded to underlying), staking/lending positions; mark-to-market; P&L from trade reconstruction.
Real-time layer	Redis pub/sub (or NATS/Kafka) fanning chain events to a WebSocket service for live prices, trades, portfolio deltas.
API & auth	REST (\$2.3 parity) + WebSocket + optional GraphQL; API keys with per-tier limits + metering; email/social + CIP-30 sign-in-with-wallet.

7.4 Suggested Stack

Frontend: Next.js (web) + React Native/Expo (mobile); TradingView Charting Library / lightweight-charts.

Backend: TypeScript (NestJS) or Go for perf-critical ingestion; Rust for DEX decoders if needed. **Data:**

Postgres, TimescaleDB/ClickHouse, Redis. **Infra:** Kubernetes; the node + indexer is the main ops cost.

KEY TECHNICAL RISKS

DEX decoder maintenance — protocols upgrade scripts; adapters need active upkeep.

Data accuracy — low-liquidity tokens produce noisy prices, needing robust outlier handling. **Node/indexer ops** — mitigate early by leaning on Blockfrost/Koios while building in-house.

Scale — real-time fan-out to many WebSocket clients.

8. The Agent Layer — A2A Trading & x402

AI agents are becoming first-class market participants on Cardano. They need the same three things humans get from Basilisk — **data, discovery, and execution** — but in machine-native form: programmatic, pay-per-use, and trustless. The Agent Layer turns Basilisk into the data and execution backbone for **agent-to-agent (A2A) trading**, monetized natively via the **x402** micropayment protocol.

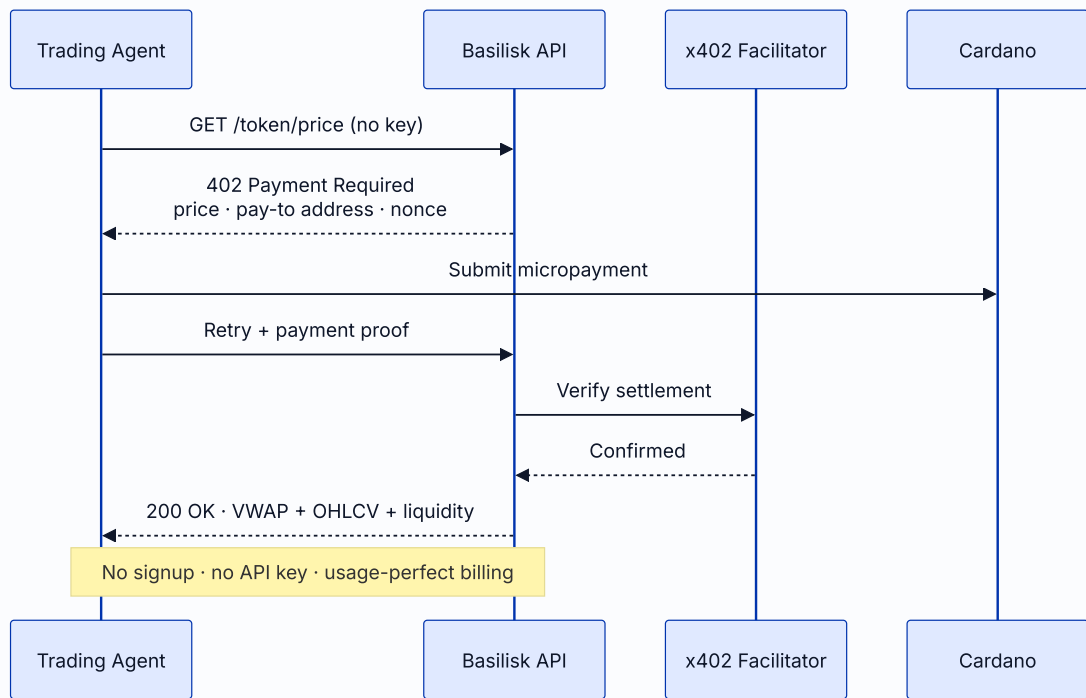
8.1 Why Agents Need Basilisk

Agents trading on Cardano today are forced through DEX UIs or raw RPC — running their own infra and stitching pricing together by hand. Basilisk already computes the hard parts: VWAP pricing, OHLCV, liquidity, smart-money signals, and aggregated swap routes. The Agent Layer simply exposes that as an agent-native interface. Critically, TapTools' API is **subscription- and API-key-gated** — friction an autonomous agent can't clear (no human to sign up for a SaaS plan with a credit card). Basilisk removes that gate entirely.

8.2 x402-Native API (Pay-Per-Call)

x402 is an HTTP-native payment standard (built on the dormant `HTTP 402 Payment Required` status, originated by Coinbase). The flow: an agent hits an endpoint, receives a `402` with payment details, settles a micro-amount on-chain, and retries with proof of payment to receive the data. **No account, no API key, no subscription.** Basilisk runs an x402 facilitator/paywall in front of the data API so every call self-settles for a fraction of a cent.

FIGURE 8.1 — X402 PAY-PER-CALL FLOW



Agents pay only for the calls they make — zero onboarding, no idle subscription cost.

This lets a **three-tier access model** coexist, each serving a different consumer:

Tier	Who	Mechanism
Free	Hobbyists, evaluation	Rate-limited public endpoints, no payment.
x402 pay-per-call Agent-native	Autonomous agents & bots	Micropayment per request; no signup, no key — settles on-chain.
Subscription	Humans, teams, enterprises	Flat monthly fiat/ADA rate with API key; predictable cost at volume.

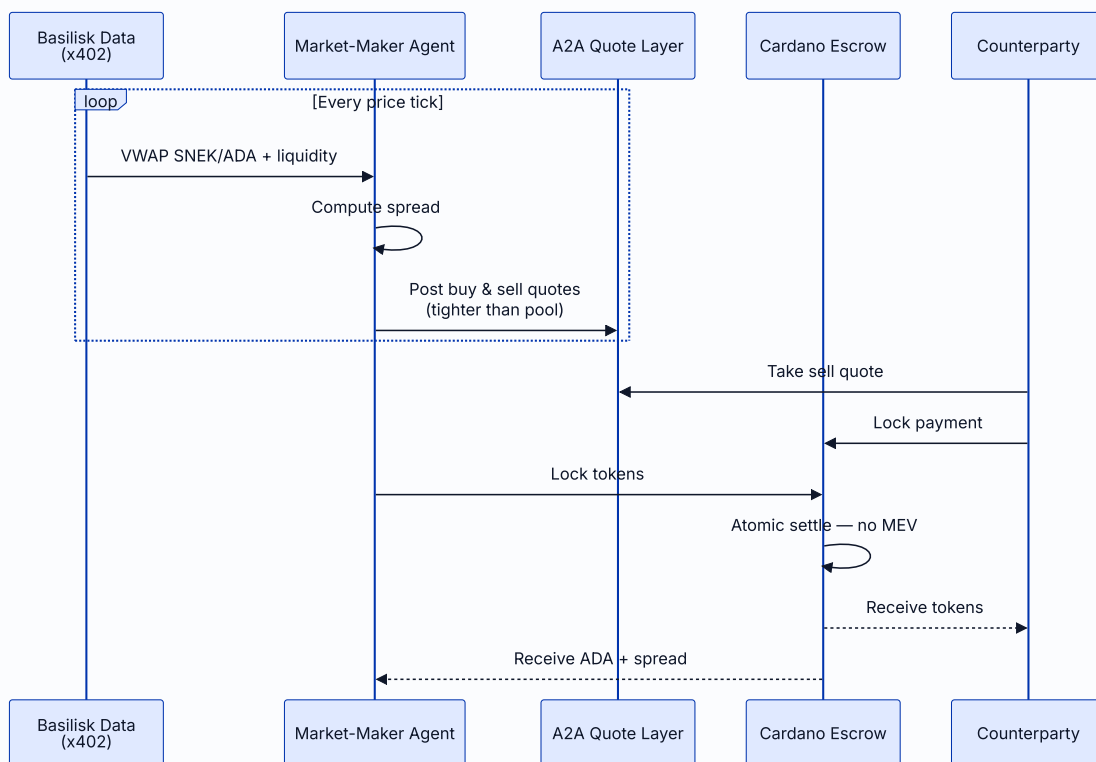
8.3 Agent-to-Agent (A2A) Trading Rails

With pricing and execution exposed, Basilisk becomes the layer agents use to trade **with each other**:

- **Price discovery** — real-time VWAP, liquidity depth, and smart-money signals via the x402 API / MCP server.
- **Execution** — the Phase 2 swap aggregator is callable programmatically; agents route and settle on-chain at best price.
- **Optional A2A quote layer** — market-maker agents post buy/sell quotes (tighter than pool spreads) that other agents or humans take, settled trustlessly via a Cardano Plutus escrow contract. Cardano's

deterministic, no-public-mempool model means **no sandwich/MEV extraction** by construction.

FIGURE 8.2 — AGENT MARKET-MAKING & A2A SETTLEMENT



Basilisk is the reference price; settlement is peer-to-peer via escrow — Basilisk never holds funds or operates a pool.

8.4 Agent Identity, Reputation & Safety

- **First-class agent wallets** — agent PnL leaderboards (extending smart-money), labeled agent clusters, and per-agent activity feeds.
- **Identity** — optional agent handles via ADA Handle (`$handle`) plus on-chain reputation accrued from settled trades.
- **Safety rails** — per-key spend caps, request rate limits, and a stale-price circuit-breaker that pauses an agent if reference data is older than a threshold.

8.5 MCP Server

An **MCP server** wraps the x402 API so LLM agents (Claude and others) can query Basilisk with natural-language and structured tools — "what's the 24h VWAP and top holders for SNEK?" — and pay per call transparently. This makes Basilisk the default Cardano data source for the agent ecosystem.

SCOPE HONESTY

The x402 API, MCP server, and agent analytics are aligned with our "aggregate, don't operate liquidity" non-goal — Basilisk serves data and routes swaps but never custodies funds. The **optional A2A quote layer (8.3)** edges toward operating a venue; it is explicitly **Phase 3 and opt-in**, settles peer-to-peer via escrow (not a pool), and is gated on the open questions in §12.

9. Monetization

Stream	Detail
Consumer subscription	Free / Pro / Trader tiers in transparent fiat + ADA. Optional NFT / \$BSL - holder perks (discount/access) — a perk, not the only path.
Day pass	Low-cost short-term access (match/beat TapTools' ~10 ADA day pass).
Developer API	Free (low rate limit) → Basic → Pro → Enterprise, metered by req/min + features (WebSocket, history depth).
Agent API (x402)	Pay-per-call micropayments from autonomous agents — usage-perfect revenue with zero onboarding (\$8.2).
Swap fees	Small bps fee on aggregated swaps, including agent-routed execution (Phase 2).
White-label / widgets	Embeds & data licensing for projects and exchanges.

PRICING PRINCIPLE

Lead with **fiat clarity** to capture the users TapTools' NFT-gating confuses. Use a native token/NFT for **loyalty perks and discounts** — never as the paywall.

10. Success Metrics (KPIs)

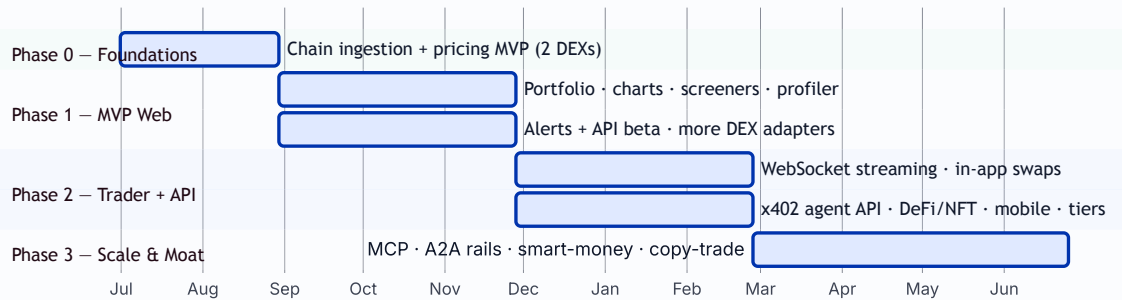
Category	Metrics
Acquisition	WAU/MAU, wallets connected/watched, sign-ups.
Engagement	Charts viewed, screeners used, alerts created, API calls/day.
Data quality	Price freshness (p95 latency), uptime, # DEXs/protocols covered, token coverage vs. TapTools.
Monetization	Paid conversion %, MRR, API revenue, ARPU.
Developer ecosystem	Active API keys, third-party apps built on Basilisk.

NORTH-STAR METRIC

Daily active wallets tracked — it combines reach and stickiness in a single number.

11. Roadmap

FIGURE 11.1 — INDICATIVE 12-MONTH PLAN



Each phase ships a usable surface; the node/indexer build de-risks early via managed fallbacks.

Phase	Timeline	Deliverables
0 — Foundations	Month 0–2	Chain ingestion (Blockfrost/Koios + Ogmios/Oura), pricing engine MVP for Minswap + SundaeSwap, Postgres/Timescale.
1 — MVP Web	Month 2–5	Portfolio, charts, screeners, wallet profiler, basic alerts, API beta; more DEX adapters.
2 — Trader + API	Month 5–8	WebSocket streaming, in-app swaps, x402 agent API beta , DeFi dashboard, NFT analytics, mobile beta, paid tiers.
3 — Scale & Moat	Month 8–12+	MCP server, A2A trading rails , smart-money/agent clustering, embeds/white-label, copy-trading, tax partners.

12. Open Questions

1. **Token/NFT:** launch a native Basilisk token/NFT (e.g. `$BSL`) for perks, or stay fiat-only at launch?
2. **Swap routing:** build our own aggregator or partner with DexHunter for execution?
3. **Indexer build vs. buy:** how long do we lean on Blockfrost/Koios before full in-house indexing?
4. **x402 settlement asset:** settle micropayments in ADA, a Cardano-native stablecoin (USDM/iUSD), or USDC on Base for cross-chain agents? Build our own facilitator or adopt an existing x402 facilitator?
5. **A2A venue boundary:** does an agent quote/offer book (§8.3) cross our "don't operate liquidity" non-goal — and if so, do we ship it as a separate product?
6. **Data licensing:** pursue B2B data feeds (exchanges, CMC/CoinGecko-style) as an early revenue line?
7. **Brand/legal:** confirm "Basilisk" name/trademark availability — it's a common mythological term, so check for crypto-name clashes and domain/handle availability.

13. TapTools Parity Checklist

- | | |
|---|---|
| ☐ Multi-wallet portfolio (ADA/USD, value-over-time, P&L) | ☐ NFT collection analytics |
| ☐ Token holdings / NFT / LP / loans | ☐ In-app swaps |
| ☐ Token price charts + OHLCV + recent trades | ☐ Alerts (price / whale / balance) |
| ☐ Top 50 / Gainers / Trending screeners | ☐ Public API (token/NFT/market/wallet) |
| ☐ Wallet Profiler (any-address deep-dive) | ☐ Day pass + subscription tiers |
| ☐ Hot wallets / smart money | ☐ WebSocket streaming (differentiator) |
| ☐ Token distribution / top holders | ☐ x402 pay-per-call API (differentiator) |
| ☐ DEX analytics (TVL / APY / liquidity) | ☐ MCP server for agents (differentiator) |
| | ☐ A2A trading rails (differentiator) |

Sources

- TapTools — [home](#) · [portfolio](#)
 - TapTools API — [docs](#) · [API subscription](#)
 - TapTools — [GitBook FAQ](#)
 - Cardano Forum — [TapTools PRO features guide](#)
 - Cardano.org — [TapTools listing](#)
 - CoinGape — [TapTools explainer](#)
 - Similarweb — [tapttools.io competitors](#)
 - Blockfrost — [Cardano API](#)
 - Coinbase — [x402 protocol](#) (HTTP-native micropayments)
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